EDUAI ACADEMIC SMART NOTES

Generated: October 29, 2026

University Curriculum Series

Introduction to Machine Learning

Section Topic: Unsupervised Learning: Clustering & Dimensionality Reduction

Executive Summary: Explores unsupervised discovery of geometric clusters with K-Means and DBSCAN, alongside spectral projection via Principal Component Analysis.

1. Core Learning Objectives

- Understand the mechanics and convergence guarantees of K-Means clustering
- Perform density-based spatial clustering (DBSCAN) for non-spherical clusters
- Derive Principal Component Analysis (PCA) using covariance matrices and eigendecomposition
- Apply dimensionality reduction for high-dimensional data compression and visualization

2. Conceptual Breakdown & Technical Formulations

1. The Core Objective of Unsupervised Learning

Without ground-truth annotations y, the algorithm must extract latent geometry, density variations, or intrinsic coordinate frames from $X \in \mathbb{R}^{N \times D}$.

2. K-Means Clustering Algorithm

Given target cluster count K, K-Means minimizes the Within-Cluster Sum of Squares (WCSS), also called inertia:

$$J = \sum_{k=1}^K \sum_{x_i \in S_k} \| x_i - \mu_k \|^2$$

Algorithm Iterations:

1. *Initialization*: Pick K initial centroids $\mu_1, \dots, \mu_K$ (preferably via K-Means++).

2. *Assignment step*: Assign each sample x_i to the nearest centroid:

$$c_i = \arg\min_k \| x_i - \mu_k \|^2$$

3. *Update step*: Recompute cluster centroids as the arithmetic mean of assigned points:

$$\mu_k = \frac{1}{|S_k|} \sum_{x_i \in S_k} x_i$$

4. Repeat until convergence.

3. Density-Based Clustering: DBSCAN

Unlike K-Means, which assumes convex, spherical clusters and requires specifying K, **DBSCAN** groups points based on density connectivity:

- Core points: At least **min_samples** within distance \epsilon.
- Border points: Within \epsilon of a core point but fewer than **min_samples**.
- Noise: Points that are neither core nor border (handles outliers automatically).

4. Principal Component Analysis (PCA)

High-dimensional datasets suffer from the *Curse of Dimensionality*. PCA finds an orthogonal coordinate system where components maximize sample variance.

1. Center data: \tilde{X} = X - \bar{X}

2. Compute sample covariance matrix:

\Sigma = \frac{1}{N-1} \tilde{X}^T \tilde{X}

3. Perform eigendecomposition:

\Sigma v_j = \lambda_j v_j

Where eigenvalues \lambda_1 \ge \lambda_2 \ge \dots \ge \lambda_D quantify the variance explained along eigenvectors v_j.

4. Project data onto the top k principal eigenvectors: Z = \tilde{X} V_k.

5. Python Example

from sklearn.decomposition import PCA

from sklearn.cluster import KMeans

pca = PCA(n_components=2)

X_reduced = pca.fit_transform(X)

print(f"Explained Variance Ratio: {pca.explained_variance_ratio_.sum():.2%}")

kmeans = KMeans(n_clusters=3, init='k-means++', random_state=42)

labels = kmeans.fit_predict(X_reduced)

6. Exam-Important Concepts

- **Elbow Method & Silhouette Analysis**: Techniques used to determine optimal K in K-Means.
- **PCA is sensitive to feature scale**: Always standardize features with **StandardScaler** before performing PCA.

3. Self-Assessment & Exam Checklist

Topic Milestone	Mastery Check
Can you formulate the primary mathematical loss / model objective?	[ ] Confirmed
Can you distinguish parametric vs non-parametric assumptions?	[ ] Confirmed
Are you prepared to calculate precision, recall, and F1 metrics?	[ ] Confirmed
Do you understand the difference between L1 sparsity and L2 shrinkage?	[ ] Confirmed